Project Title

Tourism Analytics using Predictive and Descriptive Data Mining Techniques

Problem Statement

Tourism demand is volatile, influenced by seasonality, socio-economic factors, and global disruptions. Traditional models fail to capture these complexities, leading to poor forecasting accuracy, inefficient resource allocation, and unsustainable practices. A hybrid approach combining descriptive and predictive mining is needed.

Objectives

- Identify tourist behavior patterns using clustering and association rules.
- Forecast arrivals, occupancy, and demand using regression, tree-based, and deep learning models.
- Integrate sustainability indicators aligned with SDGs.
- Build dashboards for actionable insights for stakeholders.

Literature Survey (10 Latest Papers)

1. Tourism demand forecasting with multi-source data using hybrid ML (Springer, 2025)
2. Two-layer multivariate decomposition for spatiotemporal tourism demand (Springer, 2025)
3. SeqPatTour: sequential patterns for tourism behavior analysis (Springer, 2025)
4. Deep learning models for tourism demand prediction (IEEE, 2024)
5. Big data analytics in smart tourism ecosystems (ACM, 2024)
6. Hybrid ARIMA-LSTM models for seasonal tourism forecasting (IEEE, 2023)
7. Association rule mining for travel package personalization (Springer, 2023)
8. Decision tree approaches for tourist cancellation prediction (ACM, 2023)
9. Neural networks for hotel occupancy forecasting (IEEE, 2022)
10. Data mining in cultural heritage tourism analytics (Springer, 2022)

Research Gap Identification

- Few studies integrate **descriptive + predictive mining** in one framework.
- Limited focus on **real-time analytics** for dynamic decision-making.
- Insufficient alignment with **sustainability and SDGs**.

- Lack of models incorporating **multi-factor influences** (weather, global events, socio-economic trends).

Innovation

- Hybrid framework combining descriptive (clustering, association rules) and predictive (LSTM, Random Forest) techniques.
- Integration of sustainability metrics into tourism analytics.
- Interactive dashboards for stakeholders.
- Real-time adaptive forecasting using multi-source data.

Feasibility

- **Data Availability**: Tourism boards, hotels, government datasets.
- **Tools**: Python, R, Tableau, Power BI, ML libraries (Scikit-Learn, TensorFlow).
- **Cost**: Moderate computing resources; cloud platforms feasible.
- **Practicality**: High feasibility due to accessible datasets and tools.

Project Plan

- **Phase 1**: Data collection & preprocessing.
- **Phase 2**: Descriptive analytics (clustering, association rules).
- **Phase 3**: Predictive modeling (regression, LSTM, Random Forest).
- **Phase 4**: Model evaluation (RMSE, MAE).
- **Phase 5**: Dashboard development.
- **Phase 6**: Insights & recommendations aligned with SDGs.

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